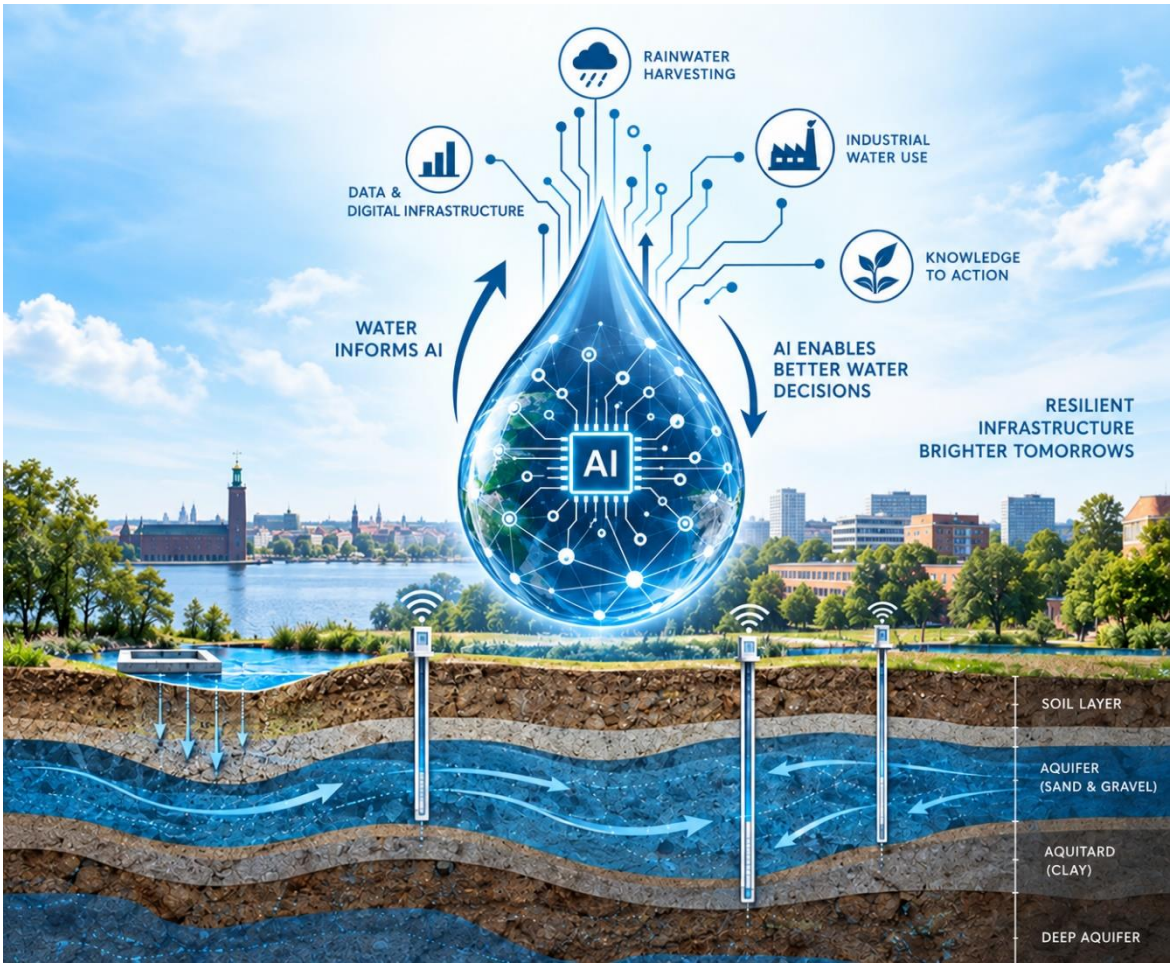
REPORT

# AI, Water and Resilient Infrastructure

## From Groundwater Intelligence to Action

**PROSUN BHATTACHARYA**




**SEMINAR REPORT**

September 2026

REPORT

# AI, Water and Resilient Infrastructure

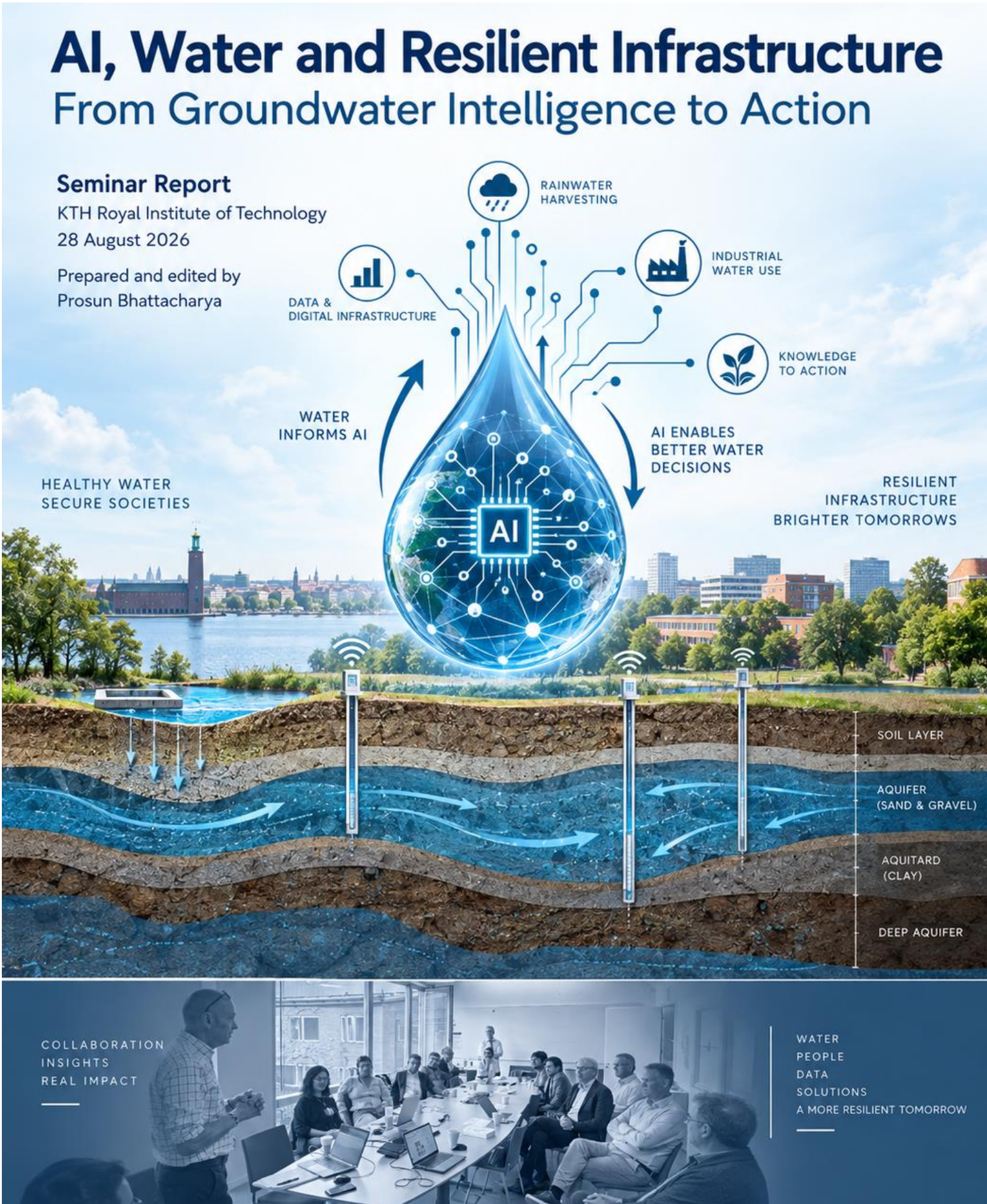
## From Groundwater Intelligence to Action

**Seminar Report**

KTH Royal Institute of Technology

28 August 2026

Prepared and edited by
Prosun Bhattacharya



**Cover illustration:** Conceptual representation of the interconnected roles of water, artificial intelligence and digital infrastructure in supporting monitoring, water management, decision-making and resilient infrastructure. The illustration integrates rainwater harvesting, industrial water use, groundwater systems, digital monitoring and AI-enabled knowledge-to-action.

This report is an edited synthesis of the seminar presentations and roundtable discussion. Individual concepts, data, illustrations and other presentation-specific material remain attributable to the respective presenters and organisations.

**Prepared and edited by**

Prosun Bhattacharya
Professor of Groundwater Chemistry & Digital Futures Faculty
KTH-International Groundwater Arsenic Research Group
Department of Sustainable Development, Environmental Science and Engineering (SEED)
KTH Royal Institute of Technology, Stockholm, Sweden

**Published by**

KTH Royal Institute of Technology
School of Architecture and the Built Environment (ABE)
Stockholm, Sweden

**Report series**

TRITA-ABE-RPT-2623

**ISBN**

978-91-8106-731-6

**Seminar date**

28 August 2026

**Suggested citation**

Bhattacharya, P. (2026). *AI, Water and Resilient Infrastructure: From Groundwater Intelligence to Action*. Seminar Report. KTH Royal Institute of Technology, Stockholm, Sweden. TRITA-ABE-RPT-2623, ISBN: 978-91-8106-731-6. 26 p.

**Disclaimer**

This report reflects the seminar discussions and related meeting material and is not a verbatim transcript. The views expressed in individual contributions remain those of the respective contributors and should not necessarily be interpreted as institutional positions of KTH Royal Institute of Technology or other participating organisations.

## Sammanfattning

KTH-seminariet den 28 augusti 2026 belyste två sidor av samma utveckling: artificiell intelligens kan stärka förståelsen och förvaltningen av vattenresurser, samtidigt som AI och den digitala infrastrukturen i sig skapar nya vattenbehov och lokala vattenrisker. Ett genomgående budskap från presentationerna och rundabordsdiskussionen var att AI inom vattenområdet får verklig betydelse först när förståelsen av de fysiska processerna kopplas samman med monitorering, tillförlitliga data, tydligt institutionellt ansvar och praktiska förutsättningar för genomförande.

**Prosun Bhattacharya** lade den vetenskapliga grunden genom att beskriva grundvatten som en till stor del osynlig samhällsresurs och infrastruktur samt betona att mer data inte automatiskt innebär bättre förståelse. Han framhöll att AI bör användas för att stärka hydrogeologisk tolkning och konceptualisering från lokal till regional och, där det är relevant, potentiellt global skala. Erfarenheter från Bangladesh visade utvecklingen från fältbaserad identifiering av säkra akviferer och lokal kunskap till digital grundvattenintelligens och realtidsmonitorering. **Joydeep Dash** kompletterade detta perspektiv genom att vända frågan från ”AI för vatten” till ”vatten för AI” och belyste den snabba utbyggnaden av datacenter i vattenstressade indiska städer samt behovet av att beakta lokal vattenrisk, transparens, cirkulär vattenanvändning, investeringar och möjligheter till samarbete mellan Sverige och Indien. **Sanjeev Sharma** presenterade en fysikinformerad neural nätverksmodell för analys av antropogen grundvattenstress i Bangladesh. Arbetet visade hur observationer av grundvattennivåer kan kombineras med akviferens massbalans, samtidigt som det tydliggjordes att de tre studerade platserna dominerades av jordbruksrelaterad vattenanvändning och att industriell påverkan behöver prövas i särskilda industriella miljöer. **Per Kinnander** presenterade WaterFORCE som en kunskap-till-handling-process där forskning, offentliga aktörer, vatten- och VA-organisationer, industri och lokalsamhällen kopplas samman genom gemensamma kunskapsunderlag, scenarioanalys, samordnade åtgärder och kontinuerligt lärande. **Md Zarif Oeishik** presenterade ett skalbart screeningkoncept för takbaserad regnvatteninsamling inom textil- och konfektionssektorn (RMG), med syfte att identifiera var regnvatteninsamling kan bidra till att minska beroendet av grundvatten innan mer detaljerade genomförbarhetsstudier genomförs.

Panel- och rundabordsdiskussionen förde samman dessa perspektiv och behandlade bland annat utformning av monitoreringssystem, identifiering av orsaker till förändringar i grundvattennivåer, behovet av litologisk information, datadelning och integritet, industriella och gruvrelaterade tillämpningar, lokal förankring och legitimitet, kvalitetssäkring samt behovet av återanvändbara och skalbara Water–AI-demonstratorer snarare än fristående pilotprojekt. AI behöver vatten – och vatten behöver i allt högre grad intelligens. Målet är inte AI för AI:s egen skull, utan att utveckla processförankrad, lokalt anpassad och ansvarsfull vattenintelligens som förkortar vägen från observation och förståelse till välgrundade beslut och ansvarsfull handling.

**Nyckelord:** Artificiell intelligens; vatten; grundvatten; vattenintelligens; digitalisering; beslutsstöd; resilient infrastruktur; hållbar vattenförvaltning

## Table of Contents

**Sammanfattning** ..... **iii**
**Foreword** ..... **vii**
**Acknowledgements** ..... **ix**
**Executive Summary** ..... **xi**
**Speakers and Contributors** ..... **xiii**
Embassy representation ..... xiv
**1. Background and Purpose** ..... **1**
1.1 Seminar objectives ..... 1
**2. Opening of the Seminar** ..... **2**
Welcome and Opening (Richard T. Olsson, CBH) ..... 2
**3. Presentation Contributions** ..... **2**
**3.1 From World Water Week to Action: The Emerging Water-AI Agenda (Prosun Bhattacharya)** **2**
3.1.1 From more data to better decisions ..... 2
3.1.2 Groundwater as hidden infrastructure: data are not understanding ..... 2
3.1.3 AI for conceptualising hydrogeological scenarios across scales ..... 2
3.1.4 From artificial intelligence to subsurface intelligence ..... 3
3.1.5 Bangladesh: from safe-aquifer targeting to digital groundwater intelligence ..... 3
**3.2 AI Needs Water: Water Risk in the Age of Digital Infrastructure (Joydeep Dash)** ..... **3**
3.2.1 AI infrastructure and local water stress ..... 3
3.2.2 Sweden-India collaboration, transparency and finance ..... 3
**3.3 Diagnosing Anthropogenic Groundwater Stress in Bangladesh with Physics-Informed Neural Networks (Sanjeev Sharma)** ..... **4**
3.3.1 What the current cases show ..... 4
3.3.2 Industrial groundwater stress as the next test case ..... 4
3.3.3 Observed limitations ..... 4
**3.4 WaterFORCE: Global Knowledge to Action (Per Kinnander)** ..... **4**
**3.5 Water-Smart Industry: Reducing Groundwater Dependence in Global Supply Chains (Md Zarif Oeishik)** ..... **5**
3.5.1 A typology-based screening concept ..... 5
3.5.2 From screening to action ..... 5
**4. Discussion Emerging During the Panel and Roundtable** ..... **6**
**4.1 Monitoring, abstraction and model credibility** ..... **6**
**4.2 Data integration, sharing and privacy** ..... **6**
**4.3 Mining, industry and critical-minerals applications** ..... **7**
**4.4 Governance, assurance and community legitimacy** ..... **7**
**4.5 From isolated models to reusable intelligence** ..... **7**
**5. Key Messages Emerging from the Seminar** ..... **8**
**6. Follow-up Directions** ..... **9**
**7. Concluding Remarks** ..... **11**
**8. References** ..... **12**
Editorial record note ..... 12

# Foreword

The seminar “AI, Water and Resilient Infrastructure: From Groundwater Intelligence to Action”, held at KTH Royal Institute of Technology on 28 August 2026, was conceived as a direct follow-on to discussions during World Water Week 2026 in Stockholm, Sweden. Those discussions highlighted both the growing potential of artificial intelligence for water management and the need to consider the water demands and wider resource implications of the digital infrastructure supporting AI.

The KTH seminar was therefore convened to take the discussion a step further: from awareness of AI-water opportunities and risks towards practical questions of groundwater intelligence, resilient infrastructure, industrial water use, monitoring, modelling and pathways from knowledge to action.

The seminar was co-organised by KTH Royal Institute of Technology and WaterAid Bangladesh, building on ongoing collaboration in groundwater and sustainable water-supply research. The programme also benefited from the wider international perspective of the International Society of Groundwater for Sustainable Development (ISGSD) and from contributions spanning research, technology, industry and governance.

This report provides an edited record of the principal presentations and the discussion that emerged during the seminar. It is intended to document the meeting, preserve its main ideas and provide a basis for continued collaboration and follow-up. I am grateful to all speakers and contributors for their engagement and for the constructive discussions that made the seminar possible.

**Prosun Bhattacharya**
Professor of Groundwater Chemistry
&
Digital Futures Faculty
KTH Royal Institute of Technology
Stockholm, Sweden
September 2026

## Acknowledgements

The seminar “AI, Water and Resilient Infrastructure: From Groundwater Intelligence to Action” was co-organised by KTH Royal Institute of Technology and WaterAid Bangladesh. The collaboration reflected the strong topical connection with ongoing KTH-WaterAid Bangladesh work on hydrogeological suitability and sustainable water-supply options in Bangladesh, together with the wider questions addressed in the seminar concerning industrial and ready-made garment (RMG) groundwater dependence, alternative water sources and resilient water management.

The editor gratefully acknowledges Björn Berggren, Head of the School of Architecture and the Built Environment (ABE), and Annika Borgenstam, Vice President for Research at KTH, for their support and encouragement in connection with the seminar. Appreciation is also extended to the ABE School and particularly to Magnus Atterfors for helping to disseminate information about the seminar through KTH webpages and communication channels.

Digital Futures is acknowledged as an interdisciplinary research environment that encourages dialogue across artificial intelligence, digitalisation and major societal challenges. The International Society of Groundwater for Sustainable Development (ISGSD) is gratefully acknowledged for contributing to the wider international groundwater and sustainable-development context of the seminar.

Special appreciation is extended to WaterAid Bangladesh for its role as co-organiser and for bringing an applied water-sector and industrial-resilience perspective to the seminar, including the contribution on water-smart industry and rooftop rainwater harvesting.

The participation of Ms Rehana Parvin, Counsellor at the Embassy of the People’s Republic of Bangladesh in Stockholm, is gratefully acknowledged. Ms Parvin attended as the Embassy’s representative, delegated on behalf of Md. Nizamul Islam and H.E. Waheeda Ahmed, Ambassador of the People’s Republic of Bangladesh to Sweden. Her participation highlighted the continuing relevance of Sweden-Bangladesh cooperation in water security, climate resilience, sustainable infrastructure and emerging digital technologies.

Finally, sincere thanks are extended to all speakers and panel contributors for sharing their expertise and for the open and constructive roundtable discussion that shaped the perspectives and follow-up directions documented in this report.

**Prosun Bhattacharya**
Professor of Groundwater Chemistry
KTH Royal Institute of Technology
Stockholm, Sweden
September 2026

## Executive Summary

The KTH seminar on 28 August 2026 connected two sides of the same development: artificial intelligence can strengthen water intelligence, while AI and digital infrastructure themselves create new water demand and local water risk. Across the presentations and roundtable discussion, a central conclusion was that Water–AI becomes useful only when physical process understanding, monitoring, trustworthy data, institutional responsibility and implementation mechanisms are connected.

**Prosun Bhattacharya** established the scientific foundation by framing groundwater as hidden infrastructure and stressing that more data do not automatically produce better understanding. He argued that AI should strengthen hydrogeological interpretation and conceptualisation from local to regional and, where appropriate, potentially global scales. Experiences from Bangladesh illustrated a progression from field-based safe-aquifer targeting and local knowledge towards digital groundwater intelligence and real-time monitoring. **Joydeep Dash** complemented this perspective by reversing the lens from "AI for water" to "water for AI", highlighting the rapid expansion of data centres in water-stressed Indian cities and the need to consider local water risk, transparency, circular water use, investment and opportunities for Sweden–India collaboration. **Sanjeev Sharma** presented a physics-informed neural-network approach to anthropogenic groundwater stress in Bangladesh, showing how groundwater-level observations can be combined with aquifer mass-balance constraints while also making clear that the three analysed sites were dominated by agriculture-related water use and that industrial attribution remains to be tested in dedicated industrial settings. **Per Kinnander** presented WaterFORCE as a knowledge-to-action pathway linking research, public authorities, utilities, industry and communities through shared evidence, scenario exploration, coordinated action and continuous learning. **Md Zarif Oeishik** presented a scalable screening concept for rooftop rainwater harvesting in the textile and ready-made garment sector, aimed at identifying where rainwater harvesting could meaningfully reduce groundwater dependence before more detailed feasibility studies are undertaken.

The panel and roundtable discussion connected these contributions to monitoring-system design, attribution of groundwater-level changes, the need for lithological information, data sharing and privacy, industrial and mining applications, community legitimacy, assurance and the development of reusable and scalable Water–AI demonstrators rather than isolated pilot projects. AI needs water, and water increasingly needs intelligence. The objective is not AI for its own sake, but process-informed, locally contextualised and accountable water intelligence that shortens the path from observation and understanding to well-founded decisions and responsible action.

**Keywords:** Artificial intelligence; water; groundwater; water intelligence; digitalisation; decision support; resilient infrastructure; sustainable water management

## Speakers and Contributors

The profiles below identify the speakers and substantive contributors to the seminar.

### Richard T. Olsson

*Professor, Polymeric Materials, KTH Royal Institute of Technology – Seminar anchor*

Richard T. Olsson is Professor of Polymeric Materials at KTH. His research spans functional polymer materials, nanoparticle-based systems and sustainable materials. He opened the seminar and helped frame its interdisciplinary focus on resilient infrastructure, technology and the translation of research into practical solutions.

### Prosun Bhattacharya

*Professor of Groundwater Chemistry, KTH Royal Institute of Technology - Seminar convenor and speaker*

Prosun Bhattacharya is Professor of Groundwater Chemistry and Coordinator of the KTH-International Groundwater Arsenic Research Group. His research focuses on groundwater quality, hydrogeochemistry, geogenic contaminants and the translation of scientific understanding into safe-water interventions, monitoring and decision support across South Asia and other regions.

### Joydeep Dash

*Founder, Open Innovators - Speaker*

Joydeep Dash is an impact entrepreneur and founder of Open Innovators with extensive experience in IT transformation, cross-border innovation and climate-focused partnerships. His seminar contribution examined the water demands and local risks associated with rapidly expanding AI and data-centre infrastructure, with a particular focus on India and Sweden-India collaboration.

### Sanjeev Sharma

*Founder & CEO, ExcelDots AB - Speaker and panel contributor*

Sanjeev Sharma is Founder and CEO of ExcelDots AB and has more than 17 years of experience in high-technology industries including semiconductors and wireless technologies. His work increasingly applies digital methods, AI/ML and connected monitoring to groundwater management and decision support.

### Per Kinnander

*Euriqon AB - Speaker and panel contributor*

Per Kinnander works with technology strategy, decision support and collaborative approaches for translating knowledge into coordinated action. At the seminar he presented the WaterFORCE perspective, emphasising shared evidence, institutional responsibility, scenario exploration, assurance and continuous learning in water-related decisions.

### Magnus Idhage

*Euriqon AB - Digital systems contributor*

Magnus Idhage works with digital platforms for scenario analysis, decision support and organisational development. His contribution supported the discussion on how multi-stakeholder information, system representations and operational decision processes can be connected without displacing institutional responsibility.

### Md Zarif Oeishik

*WaterAid Bangladesh - Speaker and panel contributor*

Md Zarif Oeishik works in the technical team of WaterAid Bangladesh on applied water, sanitation and urban resilience initiatives. His seminar presentation focused on water-smart industry and a scalable screening approach for rooftop rainwater harvesting to reduce groundwater dependence in Bangladesh's textile and RMG supply chains.

### Mattias von Brömssen

*Hydrogeologist, Sweco Sverige AB - Panel moderator*

Mattias von Brömssen is a hydrogeologist with long experience in groundwater resources, environmental assessment and water infrastructure. His doctoral work at KTH addressed arsenic-affected aquifer systems in Bangladesh. He helped anchor the panel and roundtable discussion, connecting the presentation themes to practical hydrogeological questions.

### Lizangela Sofia Huallpara Lliully

*Doctoral researcher, KTH-International Groundwater Arsenic Research Group, Department of Sustainable Development, Environmental Science and Engineering, KTH - Panel contributor*

Lizangela Sofia Huallpara Lliully is a doctoral researcher at KTH working on hydrogeochemistry and the spatial distribution and co-occurrence of geogenic contaminants, including arsenic, fluoride, boron and lithium, in volcanic and evaporitic environments of the Bolivian Altiplano.

### Maria Isabel Chambi Tapia

*Doctoral researcher, KTH-International Groundwater Arsenic Research Group, Department of Sustainable Development, Environmental Science and Engineering, KTH - Panel contributor*

Maria Isabel Chambi Tapia is a doctoral researcher at KTH with research interests in hydrogeochemistry and water-quality processes in the Bolivian Altiplano. She contributed to the seminar discussion from the perspective of field hydrogeochemistry and international groundwater research.

## Embassy representation

### Rehana Parvin

*Counsellor, Embassy of the People's Republic of Bangladesh in Stockholm - Embassy representative*

Rehana Parvin attended the seminar as the Embassy's representative, delegated on behalf of Md. Nizamul Islam and H.E. Waheeda Ahmed, Ambassador of the People's Republic of Bangladesh to Sweden. Her presence reflected the wider relevance of the seminar to Sweden-Bangladesh cooperation in water security, climate resilience and sustainable infrastructure.



*General seminar discussion at KTH ABE, Stockholm, 28 August 2026.*

# 1. Background and Purpose

The immediate impetus for the seminar emerged from discussions at World Water Week 2026 in Stockholm, Sweden, organised by the Stockholm International Water Institute (SIWI), where discussions on AI–water risks and opportunities highlighted a rapidly developing two-way relationship (SIWI, 2026). Artificial intelligence offers new capabilities for monitoring, pattern recognition, prediction, early warning and decision support, while the digital infrastructure enabling AI is itself dependent on water, energy and material resources. The KTH seminar was designed as a focused continuation of that international discussion, moving from awareness towards application (KTH Royal Institute of Technology, 2026).

A central proposition was that water consumption is not equivalent to water risk. The consequences of a withdrawal depend on source, location, season, aquifer or basin condition, water quality, competing users, ecological requirements and climate stress. At the same time, increasing volumes of environmental data do not automatically produce better decisions unless those data are interpreted through an understanding of the physical system and linked to responsible institutions and actions.

Groundwater provided a particularly relevant lens because it is both critical and largely hidden. Aquifer architecture, recharge, flow paths, groundwater-surface-water interaction and contaminant mobilisation are only partially observable, even as digital monitoring and analytical capabilities expand. The seminar therefore placed strong emphasis on the transition from data to understanding and from generic AI towards process-informed groundwater intelligence.

Bangladesh provided an important, applied context through long-standing groundwater research by the KTH-International Groundwater Arsenic Research Group and the ongoing collaboration with WaterAid Bangladesh. Within the seminar, this context was extended to industrial and RMG groundwater dependence, rooftop rainwater harvesting and the challenge of linking monitoring and modelling to practical water-management choices in Dhaka, Chattogram and other industrial settings.

## 1.1 Seminar objectives

- Translate the World Water Week 2026 AI-water debate into an applied research and action agenda.
- Connect field observations, hydrogeology, hydrochemistry, real-time monitoring and AI/ML.
- Examine how process-informed and physics-informed approaches can support groundwater interpretation and decision support.
- Explore the water demands of digital and industrial infrastructure through place-based water-risk thinking.
- Consider practical source-diversification and demand-reduction options, including rooftop rainwater harvesting for groundwater-dependent industry.
- Identify governance, data-sharing and implementation conditions required to move from knowledge to action.

# 2. Opening of the Seminar

## Welcome and Opening (Richard T. Olsson, CBH)

Richard Olsson opened the seminar by positioning water security as a systems problem spanning groundwater science, materials and sensors, digitalisation, infrastructure and governance. The opening framed the meeting as a bridge from scientific insight and technology development to implementation rather than as a technology showcase. It established the interdisciplinary premise that AI has value only when it is connected to the physical system, institutional responsibility and real decisions.

# 3. Presentation Contributions

The presentation summaries below follow the seminar sequence. Content presented by the speakers is kept distinct from questions and cross-cutting discussion that emerged later; those points are consolidated in Section 4.

## 3.1 From World Water Week to Action: The Emerging Water-AI Agenda (Prosun Bhattacharya)

The central scientific presentation developed the seminar's main intellectual pathway: from the AI-water questions emerging from World Water Week, through groundwater as hidden infrastructure and source-to-tap water safety, to geospatial intelligence, hydrogeological conceptualisation, safe-aquifer targeting, digital monitoring and a climate-resilient decision-support architecture. The argument throughout was that AI becomes useful for water only when constrained by hydrogeology, geochemistry, field evidence and the institutional context of decisions.

### 3.1.1 From more data to better decisions

- More monitoring, sensors and predictions do not automatically produce better decisions; interpretation and decision structure remain essential.
- Water use must be interpreted through local context: source, place, season, aquifer or basin condition, water quality, competing uses and climate stress.
- AI offers important capabilities in pattern recognition, prediction, early warning and decision support, but accountability remains with institutions, professionals and communities.
- The reciprocal challenge is equally important: AI itself requires water through data centres, cooling, energy systems and semiconductor supply chains.

### 3.1.2 Groundwater as hidden infrastructure: data are not understanding

Groundwater was presented as the hidden layer beneath smart-water systems. Pipes, treatment plants, sensors and dashboards may be visible, while aquifer architecture, hydrostratigraphy, recharge, groundwater-surface-water interaction and contaminant pathways remain only partly observed. Digitalisation can expose patterns, but it cannot substitute for a hydrogeological conceptual model. Examples involving arsenic and manganese in Bangladesh and a PFAS contamination context illustrated the need to connect observations to hydrogeological interpretation and public-health relevance.

### 3.1.3 AI for conceptualising hydrogeological scenarios across scales

A particularly important point was AI's potential to support hydrogeological conceptualisation beyond individual sites. By learning across geology, hydrostratigraphy, hydrochemistry, climate, Earth observation and monitoring records, AI can help recognise recurring aquifer-system types, contaminant-mobilisation environments, analogues and stress pathways at regional and potentially global scales. The scientific safeguard is equally

important: similarity in data patterns cannot be assumed to mean similarity in process, and global comparison must return to local field validation.

Geogenic contaminants such as arsenic and fluoride were used as examples of recurring but process-specific hydrogeochemical environments. This provides a possible basis for global-to-local scenario intelligence in which AI helps formulate and compare conceptual models while hydrogeology and geochemistry provide the physical controls.

### 3.1.4 From artificial intelligence to subsurface intelligence

**Subsurface intelligence**
Hydrogeology + Geochemistry + Field Knowledge + Monitoring + AI = Process-informed groundwater intelligence.
The intended progression is:
Observation -> Understanding -> Prediction -> Decision -> Action.

### 3.1.5 Bangladesh: from safe-aquifer targeting to digital groundwater intelligence

Bangladesh was used to demonstrate a long implementation pathway. Field hydrogeology under SASMIT linked safer aquifer conditions to recognisable sediment characteristics and local drilling practice. The Sediment Colour Tool translated scientific understanding into a practical field indicator, while later digital workflows extended borelog capture, sediment characterisation, mapping and decision support. The national framework discussed in the presentation brought together approximately 6.3 million tubewell records across 54 districts with geological, hydrogeological and borelog information, while real-time monitoring added the temporal dimension needed for groundwater dynamics and early warning.
The broader lesson was that digital groundwater intelligence is strongest when it grows from accumulated process understanding and field knowledge rather than beginning with an algorithm.

## 3.2 AI Needs Water: Water Risk in the Age of Digital Infrastructure (Joydeep Dash)

Joydeep Dash from Open Innovators turned the seminar lens around from AI for water to water for AI. His presentation connected digital infrastructure, enterprise architecture, investment and climate innovation to the physical limits of local water systems, with particular attention to India's rapidly expanding data-centre sector.

### 3.2.1 AI infrastructure and local water stress

The presentation highlighted the misconception of AI as a weightless "cloud" and emphasised the physical resource demands of data centres and supporting infrastructure. Using India as the main case, it cited data-centre capacity increasing from about 0.9 GW in 2023 toward roughly 2 GW by 2026 and focused attention on Mumbai, Hyderabad, Chennai and Bengaluru - cities where digital-infrastructure growth overlaps with existing water stress. The presentation used industry and public-source estimates to illustrate cooling-water demand and the risk of competition with drinking-water and agricultural needs.

The main conceptual point was that a single global water-use number cannot represent local risk. Water withdrawal, consumptive loss, return flow, source type, season, basin/aquifer condition and competing demand must all be considered when siting or financing new digital infrastructure.

### 3.2.2 Sweden-India collaboration, transparency and finance

- The presentation called for stronger water-use disclosure and clearer benchmarks for closed-loop and water-efficient cooling.

- Sweden-India innovation platforms were identified as potential vehicles for pairing Swedish cooling, circular-resource and clean-tech expertise with Indian infrastructure needs.
- Pilot deployments in water-stressed hubs were proposed as a way to connect technical solutions with measurable local outcomes.
- Blended finance and offtake-oriented collaboration were presented as mechanisms for moving proven water-smart solutions from demonstration to scale.

## 3.3 Diagnosing Anthropogenic Groundwater Stress in Bangladesh with Physics-Informed Neural Networks (Sanjeev Sharma)

Sanjeev Sharma (ExcelDots AB) presented work with Prosun Bhattacharya, Abhijit Mukherjee and Saifur Rahman on the use of physics-informed neural networks (PINNs) to interpret anthropogenic groundwater stress in Bangladesh. The approach combines observed groundwater-level data with aquifer mass-balance constraints so that the model fits observations while also inferring physically interpretable forcing terms.

### 3.3.1 What the current cases show

The presentation deliberately contrasted two national narratives: agricultural abstraction and industrial abstraction. The three analysed monitoring sites were, however, agriculture-dominated. Across the cases, the learned stress signals were interpreted primarily in relation to irrigation, aquaculture and depth-dependent water-quality preferences rather than industrial pumping. This distinction was presented explicitly as a limitation rather than as evidence against industrial groundwater stress.

### 3.3.2 Industrial groundwater stress as the next test case

The industrial textile belt around Dhaka, Gazipur, Savar and Narayanganj was presented as a well-documented but methodologically untested extension for the PINN approach because none of the current monitoring sites are located within that industrial corridor. The proposed next step is therefore a monitoring nest inside or adjacent to an industrial cluster, coupled with water-level logging and a credible proxy for industrial activity, so that agricultural and industrial forcing signatures can be compared.

### 3.3.3 Observed limitations

- No metered abstraction data were available for direct validation of the inferred forcing signal.
- The current analyses are single-site or single-nest studies; separating local pumping from broader regional exchange requires a spatial, multi-site framework.
- Lithological and aquifer information remain essential to constrain interpretation and avoid over-reading fitted parameters.
- The industrial attribution discussed in the presentation was explicitly framed as a testable hypothesis, not as a completed model result.

## 3.4 WaterFORCE: Global Knowledge to Action (Per Kinnander)

Per Kinnander from Euriqon AB addressed the institutional question that follows scientific analysis: how can evidence, responsibilities and different organisational perspectives be connected so that technically credible insights lead to coordinated action? WaterFORCE was presented as a collaborative knowledge-to-action framework linking research, public authorities, utilities, industry, communities and other actors rather than as a replacement for existing institutions.

The public-report synthesis retains the transferable principles rather than the detailed “operating system” architecture discussed during the seminar: shared evidence, visible decision criteria, assurance and traceability, scenario exploration before action, explicit institutional responsibility, and learning from measured outcomes. These ideas

complemented the groundwater and modelling presentations by addressing what must happen after a technically credible prediction is produced.

- Shared evidence should support, not bypass, established institutional responsibility.
- Decision criteria need to remain visible and capable of being revisited as evidence changes.
- Scenario comparison can help actors consider plausible consequences before commitments are made.
- Measured outcomes should feed back into both scientific understanding and future institutional decisions.

## 3.5 Water-Smart Industry: Reducing Groundwater Dependence in Global Supply Chains (Md Zarif Oeishik)

Md Zarif Oeishik of WaterAid Bangladesh presented a practical screening concept for rooftop rainwater harvesting (RWH) in the textile and ready-made garment sector. The presentation reframed the challenge from one-off factory audits towards a scalable supply-chain screening layer that can identify where more detailed assessment is most worthwhile.

### 3.5.1 A typology-based screening concept

The framework combines two simple facility characteristics - workforce size as a demand indicator and rooftop area as a rainwater-catchment indicator - to classify factories into nine typologies. The approach was developed and tested on a representative sample of textile factories in Bangladesh and was presented as a way for buyers, factories and programme partners to prioritise facilities before undertaking more expensive site-specific engineering studies.

The important conceptual result was a feasibility gradient rather than a single RWH prescription: some combinations of workforce and roof area are structurally more favourable than others. RWH was therefore proposed as a complementary source that can reduce groundwater dependence where local conditions support it, not as a universal substitute for groundwater.

### 3.5.2 From screening to action

- Dhaka and Chattogram provide relevant industrial settings because of concentrated RMG activity and groundwater dependence.
- Screening can provide an initial visibility layer across large sourcing bases and help focus detailed feasibility work on the most promising or most stressed facilities.
- Rainwater substitution also links water security to energy and climate benefits where groundwater pumping is energy intensive.
- The presentation clearly identified seasonality as the next methodological step: annual averages need to be complemented by monthly rainfall, storage and facility-demand information before design decisions are made.

# 4. Discussion Emerging During the Panel and Roundtable

After the formal presentations, the panel anchored by Mattias von Brömssen connected the individual contributions and shifted the discussion from “what was presented” to “what would be required to make these ideas operational”. The points below therefore record discussion, questions and emerging directions rather than attributing them back to the formal presentations.



*Roundtable discussion on groundwater intelligence, AI, resilient infrastructure and pathways from knowledge to action at KTH, 28 August 2026.*

## 4.1 Monitoring, abstraction and model credibility

- Industrial and digital-infrastructure groundwater risk requires site-specific monitoring that links production-well abstraction, observation wells, seasonal groundwater levels, recharge and water quality.
- Annual abstraction values can hide critical dry-season stress; seasonal metering and hydrographs are therefore more informative for operational risk.
- Agricultural, industrial and other pumping signals should be separated where possible so that groundwater decline is not attributed to the wrong driver.
- Nested monitoring can be costly, making it important to design each observation point around a specific uncertainty or management question.
- Detailed lithology and aquifer architecture were repeatedly identified as critical. A physics-informed model can only be credible if layering, connectivity and boundary conditions are represented with sufficient detail.
- Monitoring and modelling should be designed together so that new data reduce uncertainty that matters to a real decision.

## 4.2 Data integration, sharing and privacy

A major discussion thread concerned integration of groundwater observations, geological information and meteorological data. Swedish examples referred to national geological and meteorological data environments, while the Bangladesh work illustrated both the value and difficulty of linking agency data, field observations and model inputs across institutions.

- Relevant data may exist but remain difficult to access across agencies, limiting model integration and public decision support.
- Privacy and governance constraints can arise when precise well locations or water-quality information can be linked to individuals, properties or sensitive health and insurance implications.
- Water-AI therefore needs data governance, provenance, access rules, security and privacy by design rather than as a later compliance layer.
- Governed interfaces are needed to combine operational data and scientific evidence while keeping access, provenance, decision rules and responsibilities explicit.

## 4.3 Mining, industry and critical-minerals applications

Mining emerged as a concrete application for the combined monitoring-model-governance framework. The discussion considered how monitoring wells around an industrial or mining site could be linked with groundwater flow, water quality, landscape information and regional datasets to estimate contaminant transport, dilution time and ecological risk.

- Such a demonstrator could connect source-pathway-receptor understanding with scenario analysis and communication to affected communities.
- Critical-mineral investment increases the importance of early groundwater intelligence, environmental assurance and local legitimacy.
- The wider proposition was that water intelligence should guide infrastructure and investment before environmental conflict emerges, not only after impacts are detected.

## 4.4 Governance, assurance and community legitimacy

- AI cannot carry accountability. Public authorities, professionals, companies, utilities and communities remain responsible for decisions.
- A technically strong model can still fail if the underlying data are not trusted, the rules are unclear or affected communities do not consider the process legitimate.
- The WaterFORCE discussion provided one reference point for making evidence, decision rules, assurance and institutional roles more visible rather than leaving them implicit inside an algorithm.
- Community acceptance should be treated as an operational variable: infrastructure and abstraction projects can stall if local water consequences are not understood, communicated and addressed early.
- The discussion therefore moved from “Which model is best?” toward “Which decision architecture allows different actors to use models responsibly?”

## 4.5 From isolated models to reusable intelligence

- PINNs and hybrid models should be developed as reusable components linked to hydrogeological conceptual scenarios rather than as disconnected boxes for individual sites.
- AI can compare plausible futures under altered pumping, recharge, climate, industrial growth and land use, but uncertainty and physical constraints must remain visible.
- The global-to-local hydrogeological concept offers a possible library of process-based scenario types that can be refined with local monitoring and field knowledge.
- The discussion suggested a common demonstrator logic:

  **Observe -> Understand -> Predict -> Decide -> Act -> Measure & Learn**

  with governance and community participation embedded across the cycle.

# 5. Key Messages Emerging from the Seminar

- Water use is not the same as water risk. Place, source, timing, hydrological condition, water quality and competing demand determine the significance of a withdrawal.
- Data are not the same as understanding. Groundwater AI must remain anchored in hydrogeological and geochemical process knowledge.
- AI has potential not only for prediction but also for comparing hydrogeological scenarios and analogues across scales, provided that field validation remains the scientific control.
- Process-informed and physics-informed approaches offer a pathway between purely data-driven models and data-hungry conventional calibration, but they do not remove the need for lithology, monitoring and physical interpretation.
- Digital and industrial infrastructure should be assessed through local water-risk intelligence before siting, financing or expansion decisions are made.
- Prediction must connect to intervention. Safe-aquifer targeting, abstraction management, rainwater harvesting, reuse and monitoring upgrades are examples of actions that can follow from better intelligence.
- Governance, transparency, assurance and community legitimacy are part of the technical system. AI supports decisions; it does not inherit responsibility for them.

**Integrated seminar logic**
Observe -> Understand -> Predict -> Decide -> Act -> Learn. Hydrogeology, hydrochemistry, climate, infrastructure, institutions, communities and accountability surround the entire cycle.

# 6. Follow-up Directions

The seminar highlighted several opportunities for developing the discussion further through research, demonstration and collaboration. The directions below (Table 1) represent an editorial synthesis of the presentations and roundtable discussion and should be regarded as potential avenues for follow-up rather than formally agreed commitments or implementation timelines.

**Table 1.** Potential follow-up directions from the seminar toward Subsurface Intelligence research, demonstration and implementation

| Emerging direction | Potential follow-up |
|---|---|
| **Groundwater monitoring in industrial settings** | Explore monitoring cases in groundwater-dependent industrial areas, combining multi-depth groundwater-level observations with seasonality, abstraction pressures, water-quality information and relevant operational data. |
| **Process-informed data framework for AI applications** | Define the minimum combination of geological, hydrogeological, lithological, hydrochemical, meteorological and monitoring information required for scientifically credible AI/ML-supported groundwater interpretation and prediction. |
| **Physics-informed assessment of groundwater stress** | Test physics-informed approaches under different abstraction regimes, including industrial settings, and examine how climatic, agricultural and industrial groundwater-use signals can be better differentiated. |
| **Water-smart industry and rainwater harvesting** | Build on the RMG screening approach to identify suitable facilities for more detailed assessment of rooftop rainwater harvesting, including rainfall seasonality, storage requirements, water demand and site-specific feasibility. |
| **Global-to-local groundwater intelligence** | Further develop AI-supported regional and global hydrogeological conceptualisation while retaining local geological, hydrogeological and geochemical process understanding as the basis for interpretation and validation. |
| **Subsurface intelligence for contaminated sites** | Explore selected Swedish contaminated sites, including sites identified through the **MIFO framework**, as possible implementation cases integrating historical site information, geology, hydrogeology, hydrochemistry, monitoring and spatial data. The objective would be to improve understanding of **contaminant occurrence, subsurface migration, exposure pathways, risk and remediation performance**. |
| **From evidence to intervention** | Examine how monitoring, modelling and scientific evidence can be connected more effectively with institutional responsibilities, stakeholder engagement, assurance, traceability and adaptive management, consistent with the WaterFORCE **Knowledge → Action → Impact** perspective. |
| **Water risk associated with digital infrastructure** | Develop site-specific approaches for assessing the water implications of AI and digital infrastructure by considering groundwater dependence, local water availability, aquifer vulnerability, competing demands and basin-scale water stress. |
| **Integrated Water–AI and Subsurface Intelligence demonstrators** | Identify real-world cases where monitoring, process understanding, AI/ML and intervention can be tested together, providing a basis for interdisciplinary research, technology development and implementation. |

Taken together, these directions point towards a broader Subsurface Intelligence capability in which geological, hydrogeological, hydrochemical, geospatial, monitoring and operational information can be integrated with process-based modelling and AI/ML to support the sequence:

**Observe → Understand → Predict → Decide → Intervene → Monitor → Learn**

A longer-term opportunity is therefore to explore the establishment of a Subsurface Intelligence Hub linking research, innovation and implementation. Such a Hub could provide an environment for developing and testing monitoring technologies, process- and physics-informed AI, contaminant and remediation intelligence, groundwater decision-support systems and water-risk assessment tools around real-world demonstration cases.

Within this framework, WaterFORCE could provide an important pathway from scientific and digital intelligence to implementation, while partnerships with public authorities, consultants, technology developers and industry could create opportunities for innovation, demonstration and future business development. The Hub concept would thus extend the seminar discussion from the application of AI to individual water problems towards an integrated capability for understanding the subsurface and converting that understanding into accountable action.

# 7. Concluding Remarks

The seminar demonstrated that the future of water management is not simply digital. It must be process-informed, risk-based, place-specific and decision-oriented. AI can strengthen our ability to integrate observations, recognise patterns, explore scenarios and anticipate change, but meaningful intervention still depends on geological and hydrogeological understanding, field evidence, professional judgement, institutions and communities.

The discussions also pointed beyond individual applications of AI toward a broader concept of Subsurface Intelligence: integrating geology, hydrogeology, hydrochemistry, monitoring, spatial information and process-based modelling with AI/ML to move systematically from observation and understanding to prediction, intervention and learning. Potential applications range from groundwater security and industrial water management to contaminant migration, risk assessment and remediation at contaminated sites.

Real-world demonstrators—including industrial groundwater systems, rainwater-harvesting applications and potentially selected Swedish contaminated sites—could provide environments in which these approaches are tested and refined. In this context, the WaterFORCE Knowledge → Action → Impact perspective provides a pathway for connecting scientific and digital intelligence with implementation, governance and measurable outcomes.

The seminar therefore provides a basis for exploring a future Subsurface Intelligence Hub linking research, innovation and implementation. Such a Hub could bring together academia, public authorities, technology developers, consultants and industry around a common objective: to transform increasingly rich subsurface data into scientifically credible, actionable intelligence for sustainable and resilient water and environmental management.

AI needs water — and water increasingly needs intelligence. The next challenge is to ensure that this intelligence leads to informed and accountable action. From World Water Week to the KTH seminar, and from Subsurface Intelligence and real-world demonstrators to WaterFORCE, the emerging trajectory provides a basis for exploring a future Subsurface Intelligence Hub at KTH.



*Participants during the KTH seminar roundtable, 28 August 2026.*

# 8. References

This report was prepared from the seminar concept/programme, presenter-provided materials, discussion notes and the consolidated post-seminar record. The main seminar presentation sources were:


- Stockholm International Water Institute (SIWI) (2026). AI – Water Risks and Opportunities. World Water Week 2026, Stockholm, Sweden, 24 August 2026. Session 12738. URL: SIWI – World Water Week 2026 session (accessed 24 August 2026)
- KTH Royal Institute of Technology (2026). Seminar "AI, Water and Resilient Infrastructure – From Groundwater Intelligence to Action". URL: https://www.kth.se/en/om/2.266/seminar-ai-water-and-resilient-infrastructure-from-groundwater-intelligence-to-action-1.1490821
- Bhattacharya, P. (2026) "From World Water Week to Action: The Emerging Water-AI Agenda", KTH seminar presentation, 28 August 2026.
- Dash, J. (2026) AI-water seminar presentation, Open Innovators, KTH, 28 August 2026.
- Sharma, S., Bhattacharya, P., Mukherjee, A. and Rahman, S. (2026) Diagnosing Anthropogenic Groundwater Stress in Bangladesh with Physics-Informed Neural Networks. KTH seminar presentation, 28 August 2026.
- Kinnander, P. (2026) "WaterFORCE: Global Knowledge to Action", KTH seminar presentation, 28 August 2026.
- Oeishik, M. Z. (2026) "Water-Smart Industry: Reducing Groundwater Dependence in Global Supply Chains", WaterAid Bangladesh, KTH seminar presentation, 28 August 2026.
- Consolidated panel and roundtable discussion notes from the seminar, 28 August 2026.
- General seminar photographs reproduced with attribution to the respective photographer/rights holder where applicable.


## Editorial record note

The DiVA/TRITA version is intentionally a Seminar Report rather than a proceedings volume. It does not reproduce individual presentation cover pages, detailed unpublished datasets, confidential figures or proprietary system architecture. The purpose is to document the meeting accurately while preserving attribution and allowing presentation-specific research to be developed independently in subsequent scholarly publications.

TRITA-ABE-RPT-2623

ISBN: 978-91-8106-731-6

Stockholm, Sweden 2026

www.kth.se